A Robust Inverse Uncertainty Principle for the Boolean Cube, Formalized in Lean

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August 2026

Abstract

An uncertainty principle says that a state cannot be sharply concentrated in two incompatible descriptions at once. On n bits, the two descriptions considered here are the ordinary basis and the Walsh–Hadamard basis. The exact support uncertainty inequality says that the numbers of nonzero coordinates in these two bases have product at least 2^n. Its equality cases are uniform superpositions over shifted linear subspaces, with a possible sign pattern. We prove a quantitative inverse statement that allows small probability leakage outside prescribed sets S and T. If the leakage is at most ε and $|S||T| \leq (1+\delta)2^n$, then, in a local parameter range, both sets are close to complementary shifted subspaces and the state is close to the corresponding coset state. The dependence is explicit:

$$q = \max\{(1024\varepsilon)^{1/4},\, 4\sqrt{\delta}\},$$

with relative set errors at most $5q$ and $12q$, and squared Euclidean state error at most $264q$. The constants are not optimized. The complete proof is formalized in Lean 4 against Mathlib, with a small Mathlib-only public specification separated from the implementation.

1 Introduction

The word "uncertainty" has several mathematical meanings. The common idea is that a vector can be simple in one coordinate system or in an incompatible coordinate system, but generally not in both. For an n-qubit state, the two coordinate systems in this note are:

- the *computational basis*, labelled by n-bit strings; and
- the *Hadamard basis*, obtained by applying a Hadamard gate to every qubit.

The change of coordinates between them is the Walsh Fourier transform.

The *support* of a function is simply the set of inputs where it is nonzero. A finite version of the Donoho–Stark support uncertainty principle says that every nonzero function f on a finite abelian group G satisfies

$$|\operatorname{supp} f|\,|\operatorname{supp} \widehat{f}| \geq |G|.$$

The hat means "take the Fourier transform." Equality has a rigid form: up to harmless symmetries, f is constant on a shifted subgroup and zero elsewhere. On bit strings, subgroups are linear subspaces and shifted subgroups are affine subspaces, also called cosets.

The exact support is unstable under noise. An arbitrarily small perturbation can give a vector full support in both bases. For quantum applications it is more natural to specify sets S and T that capture almost all measurement probability. The question addressed here is:

If S and T nearly attain the support-product lower bound and a state is almost concentrated on them in complementary bases, must the sets and state be close to a hidden-coset configuration?

For the Boolean cube the answer is yes, quantitatively. The result below separates the two sources of error: ε measures probability leakage and δ measures excess in the support product. This separation is useful both mathematically and for reading the formal statement.

The qualitative phenomenon is not new. Exact uncertainty and its equality cases have an extensive literature [DS89, Mes06, BG13, FHX19, WW21]. A recent result of Miller gives a robust coefficient inequality for exact supports in arbitrary finite abelian groups [Mil26]. Section 7 gives a precise comparison. The contribution here is a particular explicit package for $\mathbb{F}_2^n$: approximate concentration sets, symmetric-difference control of both sets, complex state-vector control including phase, and a machine-checked proof. Section 6 compares this result with continuous Heisenberg stability, where the corresponding rigid objects are Gaussian wave packets rather than cosets.

2 The result in plain language

An element of $\mathbb{F}_2^n$ is an n-bit string. Addition in $\mathbb{F}_2^n$ is bitwise exclusive-or. A *linear subspace* H is a collection of bit strings that contains the all-zero string and is closed under this addition. A *coset* $a + H$ is what one gets by adding a fixed string a to every element of H.

A normalized function $f : \mathbb{F}_2^n \to \mathbb{C}$ can be read as a quantum state: $|f(x)|^2$ is the probability of observing x in the computational basis. After a Hadamard transform, the probability of observing y is $|\widehat{f}(y)|^2/2^n$. Thus the assumptions of the theorem say:

1. computational-basis measurement lands in a set S with probability at least $1 - \varepsilon$;
2. Hadamard-basis measurement lands in a set T with probability at least $1 - \varepsilon$; and
3. the product $|S||T|$ is only a factor $1 + \delta$ above the smallest value permitted by uncertainty.

The conclusion reconstructs a hidden linear structure. It finds a subspace H such that S is close to one shifted copy of H, while T is close to a shifted copy of the complementary space $H^\perp$. Here $H^\perp$ consists of the bit strings having dot product zero with every element of H. The theorem also says that the entire complex state f, not just the two sets, is close to

$$\frac{1}{\sqrt{|H|}} \sum_{x \in a+H} (-1)^{b \cdot x} |x\rangle,$$

up to one physically irrelevant global phase.

There are three numerical parameters because they play different roles:

- ε is lost probability mass;
- δ is excess in the set-size product; and
- q is the accuracy promised by the conclusion.

The theorem is local: it assumes $0 < q \leq 1/100$ and applies when

$$\varepsilon \leq q^4/1024, \qquad \delta \leq q^2/16.$$

Equivalently, one can compute q from the two observed errors by taking

$$q = \max\{(1024\varepsilon)^{1/4}, 4\sqrt{\delta}\}.$$

The fourth root means that the present quantitative bound becomes useful only when the leakage is quite small. This is a limitation of the proved estimate, not a claim that fourth-root behavior is optimal.

3 Formal statement of the result

3.1 Normalization

Fix

$$V = \mathbb{F}_2^n, \qquad N = |V| = 2^n.$$

For $f : V \to \mathbb{C}$ use the unnormalized Walsh transform

$$\widehat{f}(y) = \sum_{x \in V} (-1)^{x \cdot y} f(x).$$

Thus Parseval's identity is

$$\sum_{y \in V} |\widehat{f}(y)|^2 = N \sum_{x \in V} |f(x)|^2.$$

For $A \subseteq V$, write

$$\mathcal{E}_A(f) = \sum_{x \in A} |f(x)|^2.$$

The symmetric difference of two sets is denoted by $\triangle$.

For a subspace $H \leq V$, offsets $a, b \in V$, and a phase $c \in \mathbb{C}$ with $|c| = 1$, define the normalized modulated coset state

$$\phi_{H,a,b,c}(x) = \frac{c(-1)^{b \cdot x}}{\sqrt{|H|}} \mathbf{1}_{a+H}(x). \tag{1}$$

Its Walsh transform is supported on $b + H^{\perp}$.

3.2 Two-error theorem

Theorem 3.1 (Robust inverse uncertainty)**.** *Let $S, T \subseteq V$ be nonempty, let $f : V \to \mathbb{C}$, and let $\varepsilon, \delta, q \in \mathbb{R}$ satisfy*

$$\varepsilon, \delta \geq 0, \qquad 0 < q \leq \frac{1}{100}, \qquad \varepsilon \leq \frac{q^4}{1024}, \qquad \delta \leq \frac{q^2}{16}.$$

Suppose

$$\mathcal{E}_V(f) = 1, \tag{2}$$
$$\mathcal{E}_S(f) \geq 1 - \varepsilon, \tag{3}$$
$$\mathcal{E}_T(\widehat{f}) \geq (1 - \varepsilon)N, \tag{4}$$
$$|S|\,|T| \leq (1 + \delta)N. \tag{5}$$

Then there are a subspace $H \leq V$, offsets $a, b \in V$, and a phase $c \in \mathbb{C}$, $|c| = 1$, such that

$$|S \triangle (a + H)| \leq 5q|S|, \tag{6}$$

$$|T \triangle (b + H^{\perp})| \leq 12q|T|, \tag{7}$$

$$\sum_{x \in V} |f(x) - \phi_{H,a,b,c}(x)|^2 \leq 264q. \tag{8}$$

The role of q is only to make the conclusion and the admissible error region easy to read. It can be eliminated.

Corollary 3.2 (Independent-error form)**.** *Under* (2)–(5)*, put*

$$\rho(\varepsilon, \delta) = \max\{(1024\varepsilon)^{1/4},\, 4\sqrt{\delta}\}.$$

If $0 < \rho(\varepsilon, \delta) \leq 1/100$*, then* (6)–(8) *hold with* $q = \rho(\varepsilon, \delta)$*.*

Proof. The definition gives $\varepsilon \leq \rho^4/1024$ and $\delta \leq \rho^2/16$, so Theorem 3.1 applies. □

Remark 3.3 (Local and non-sharp constants)*.* The condition $\rho \leq 1/100$ makes this a local stability theorem. The constants 5, 12, and 264, and the powers and numerical factors in ρ, are outputs of a proof designed for transparency and formalization; no optimality is claimed. The endpoint $\varepsilon = \delta = 0$ is covered by the classical exact equality theorem. It is separated from Corollary 3.2 because that corollary uses a strictly positive control scale.

4 Proof architecture

This section gives a conventional account of the proof. The formal development cited in Section 8 contains all finite-sum, rounding, and constant calculations. A reader interested mainly in the statement or the quantum interpretation can skip this section.

4.1 The proof in five steps

The argument can be summarized without the constants as follows.

1. Restrict f to S. The small amount removed in this operation remains small after the Walsh transform by Parseval's identity, which says that the transform preserves total squared length up to the known factor 2^n.

2. Near equality in the support bound makes the submatrix of Walsh signs with rows in T and columns in S close to a rank-one matrix. A rank-one matrix is one whose rows are all scalar multiples of a single row.

3. A four-corner identity for Walsh signs converts this approximate rank-one property into a statement that most translated pairs from $S \times T$ have dot product zero.

4. An additive-combinatorial argument shows that two nearly maximally sized, nearly orthogonal sets must be close to complementary subspaces.

5. Projecting onto the two resulting cosets leaves little error. The common intersection of the two projector ranges is a single coset-state line, so f is close to a unit vector on that line.

4.2 The restricted inverse proposition

Let $P_S f = f\mathbf{1}_S$. The structural core of the proof is the following restricted-Walsh statement.

Proposition 4.1 (Restricted inverse step)**.** *Let $0 < q \le 1/100$, let $\|f\|_2 = 1$, and suppose*

$$\mathcal{E}_S(f) \ge 1 - q, \tag{9}$$

$$\mathcal{E}_T(\widehat{P_S f}) \ge \left(1 - \frac{q^2}{16}\right) N\, \mathcal{E}_S(f), \tag{10}$$

$$|S|\,|T| \le \left(1 + \frac{q^2}{16}\right) N. \tag{11}$$

Then there are H, a, b satisfying (6) and (7). Moreover, if $R_{H,a,b}$ denotes orthogonal projection onto the line spanned by

$$x \longmapsto |H|^{-1/2}(-1)^{b\cdot x}\mathbf{1}_{a+H}(x),$$

then

$$\|f - R_{H,a,b} f\|_2^2 \le 132q. \tag{12}$$

Here is the mechanism behind Proposition 4.1. Restrict the Walsh matrix to rows T and columns S. Splitting $P_S f$ into real and imaginary parts shows that one normalized real component $v : S \to \mathbb{R}$ retains the ratio in (10). An exact Frobenius identity then bounds the residual between the restricted sign matrix $((-1)^{s\cdot t})_{t,s}$ and a rank-one matrix formed from v and its restricted Walsh transform:

$$\mathrm{Res} \le \frac{q^2}{8} N.$$

Rounding the two factors to signs costs at least one unit of squared residual at every mismatching matrix entry. Hence the number of sign mismatches is at most $q^2 N/8$.

The four-point identity

$$(-1)^{s\cdot t}(-1)^{s_0\cdot t}(-1)^{s\cdot t_0}(-1)^{s_0\cdot t_0} = (-1)^{(s+s_0)\cdot(t+t_0)}$$

turns those entry mismatches into almost orthogonality. Averaging over anchors gives $s_0 \in S$ and $t_0 \in T$ such that, for

$$A = S + s_0, \qquad B = T + t_0,$$

at most $q^2|A||B|$ pairs $(a, b) \in A \times B$ have $a \cdot b = 1$, while

$$|A||B| \ge (1 - q)N.$$

The following finite additive lemma is the structural engine.

Lemma 4.2 (Saturated almost orthogonality)**.** *Suppose $A, B \subseteq V$, $A \ne \varnothing$, $0 \in B$,*

$$|A||B| \ge (1 - q)N,$$

and at most $q^2|A||B|$ pairs in $A \times B$ are nonorthogonal. If $0 < q \le 1/100$, then some subspace $L \le V$ satisfies

$$|A \triangle L^\perp| \le 5q|A|, \qquad |B \triangle L| \le 12q|B|.$$

For completeness, the lemma is obtained by taking

$$X = \{b \in B : \ |\{a \in A : a \cdot b = 1\}| \le q|A|\}.$$

Finite Markov gives $|X| \ge (1-q)|B|$. The inequality $\mathrm{bad}(x+x') \le \mathrm{bad}(x)+\mathrm{bad}(x')$, Walsh Parseval, and the saturation bound imply $|X+X| < \frac{3}{2}|X|$. The sharp small-doubling fact at the threshold $3/2$ therefore identifies $X+X$ with the point set of a subspace L. Parseval and $|L||L^\perp| = N$ give

$$|L| \le (1+10q)|B|, \qquad |L^\perp| \le (1+3q)|A|.$$

Combining these size bounds with $X \subseteq B \cap L$ and counting nonorthogonal pairs outside $L^\perp$ yields the two asserted symmetric differences.

Translating back gives the set conclusions in Proposition 4.1. To recover the vector, let P be the coordinate projector onto $a+H$ and let Q be the Fourier-coordinate projector onto $b+H^\perp$. Flatness estimates from the same restricted Walsh residual give

$$\|f - Pf\|_2^2 \le 12q, \qquad \|f - Qf\|_2^2 \le 54q.$$

The projectors commute, and their common range is the one-dimensional line in Proposition 4.1. Consequently,

$$\|f - PQf\|_2^2 \le 2\|f - Pf\|_2^2 + 2\|f - Qf\|_2^2 \le 132q,$$

which proves (12).

4.3 From two-sided concentration to the restricted hypothesis

Proof of Theorem 3.1. It is enough to use the weaker bounds

$$\mathcal{E}_S(f) \ge 1-\eta, \qquad \mathcal{E}_T(\widehat{f}) \ge (1-\eta)N, \qquad |S||T| \le (1+\theta)N,$$

where

$$\eta = \frac{q^4}{1024}, \qquad \theta = \frac{q^2}{16}.$$

Set

$$G = \mathcal{E}_S(f), \qquad A = \mathcal{E}_T(\widehat{P_S f}), \qquad e = f - P_S f.$$

Then $\|e\|_2^2 = 1 - G \le \eta$, so Parseval gives $\|\widehat{e}\|_2^2 \le \eta N$. For every $r > 0$, the pointwise Young inequality

$$|z+w|^2 \le (1+r)|z|^2 + (1+r^{-1})|w|^2$$

gives

$$\mathcal{E}_T(\widehat{f}) \le (1+r)A + (1+r^{-1})\eta N.$$

Choose $r = q^2/32 = \theta/2$. The exact budget identity

$$(1+r)(1-\theta) + (1+r^{-1})\eta = 1-\eta$$

and the bound $G \le 1$ show that the right side of Young's inequality would be at most $(1-\eta)N$ if A were at most $(1-\theta)NG$. Comparing with the lower bound on $\mathcal{E}_T(\widehat{f})$ therefore forces

$$A \ge (1-\theta)NG.$$

Also $\eta \le q$, so $G \ge 1-q$. Thus all hypotheses of Proposition 4.1 hold, and it supplies H, a, b and the rank-one residual (12).

Let v be the normalized vector spanning that line and put $d = \langle v, f\rangle$. Orthogonal projection gives

$$\|f - dv\|_2^2 = 1 - |d|^2.$$

Choose $c = d/|d|$ when $d \neq 0$, and $c = 1$ otherwise. Since $|d| \leq 1$,

$$\|f - cv\|_2^2 = 2 - 2|d| \leq 2(1 - |d|^2) = 2\|f - dv\|_2^2 \leq 264q.$$

The vector cv is exactly (1), completing the proof. $\square$

5 Quantum interpretation

Regard f as the computational-basis amplitudes of an n-qubit pure state

$$|\psi\rangle = \sum_{x \in V} f(x)|x\rangle.$$

Equation (3) says that computational-basis measurement lands in S with probability at least $1 - \varepsilon$. Because the amplitudes of the state after $H^{\otimes n}$ are $\widehat{f}(y)/\sqrt{N}$, equation (4) says that Hadamard-basis measurement lands in T with the same probability.

The state (1) is an affine CSS stabilizer state. For $h \in H$, translation X^h has eigenvalue $(-1)^{b \cdot h}$; for $k \in H^\perp$, the phase operator Z^k has eigenvalue $(-1)^{k \cdot a}$. Put less formally, "CSS stabilizer state" here means a uniform superposition over solutions of linear equations in the bits, with a linear pattern of plus and minus signs. The theorem therefore says that near-minimal concentration in two complementary bases certifies proximity to a hidden CSS/coset state, not merely that the two concentration sets look affine.

Let $|\phi\rangle$ denote the state in (1). For pure states,

$$D_{\mathrm{tr}}(\psi, \phi) = \sqrt{1 - |\langle\psi|\phi\rangle|^2} \leq \||\psi\rangle - |\phi\rangle\|_2 \leq \sqrt{264q}.$$

Equivalently, the infidelity is at most $264q$. Every measurement outcome distribution for the two states is therefore within total variation distance $\sqrt{264q}$. These bounds are operationally meaningful but numerically coarse; improving the state constant is a natural next target.

6 Comparison with the continuous theorem

It is useful to compare Theorem 3.1 with the familiar uncertainty principle for a particle moving on the real line. This comparison explains what kind of result we have proved, but the two theorems are not special cases of one another.

Let $\psi : \mathbb{R} \to \mathbb{C}$ be a normalized wavefunction. Its position distribution is $|\psi(x)|^2$, and its momentum distribution is the squared magnitude of its Fourier transform. Write σ_x and σ_ξ for their standard deviations. With the normalization

$$\widehat{\psi}(\xi) = (2\pi)^{-1/2} \int_{\mathbb{R}} e^{-ix\xi} \psi(x)\, dx,$$

the Heisenberg inequality is

$$\sigma_x \sigma_\xi \geq \frac{1}{2}. \tag{13}$$

Equality holds exactly for Gaussian wave packets, after allowing translation, modulation, dilation, and global phase [FS97].

There is also a robust inverse statement: if the left side of (13) is close to 1/2, then the wavefunction is close to a Gaussian after applying those symmetries. The following elementary one-dimensional version makes this precise.

Proposition 6.1 (Continuous Gaussian stability)**.** *Suppose $\|\psi\|_2 = 1$, its position and momentum variances are finite, and*

$$\sigma_x \sigma_\xi \leq \frac{1+\kappa}{2}$$

for $\kappa \geq 0$. Then there are $x_0, \xi_0, \theta \in \mathbb{R}$ and $s > 0$ such that the normalized Gaussian wave packet

$$g(x) = e^{i\theta} e^{i\xi_0 x} s^{1/2} \pi^{-1/4} e^{-s^2(x-x_0)^2/2}$$

satisfies

$$\|\psi - g\|_2^2 \leq \kappa.$$

Proof. Translate and modulate ψ so that the two means are zero, then dilate it so that the position and momentum variances are equal. These operations do not change their product. For the resulting function u,

$$\langle u, (-d^2/dx^2 + x^2)u \rangle = 2\sigma_x \sigma_\xi \leq 1 + \kappa.$$

The harmonic oscillator $-d^2/dx^2 + x^2$ has a Gaussian ground state with eigenvalue 1, and every orthogonal state has energy at least 3. This spectral gap implies that the squared component of u perpendicular to the Gaussian is at most $\kappa/2$. If α is the overlap with the normalized ground-state Gaussian, choosing the best global phase gives

$$\|u - e^{i \arg \alpha} g_0\|_2^2 = 2 - 2|\alpha| \leq 2(1 - |\alpha|^2) \leq \kappa.$$

Undoing the translation, modulation, and dilation produces the required Gaussian packet. □

The proposition is a standard spectral-gap argument; much sharper and higher-order stability results are part of an active literature [DLL26]. The analogy with our finite theorem is:

	Continuous Heisenberg	Boolean support theorem
Space	Real line	n-bit strings
Measure of spread	Variance	Concentration-set size
Exact minimizers	Gaussian packets	Modulated coset states
Main rigidity tool	Harmonic-oscillator gap	Additive combinatorics
Robust conclusion	Close to a Gaussian	Close to a coset state

The shared principle is an *inverse theorem*: near equality in an uncertainty inequality forces closeness to the family of exact equality cases. The important difference is the geometry. Variance and smoothness on $\mathbb{R}$ lead to Gaussians and a linear spectral argument. Cardinality and XOR-addition on $\mathbb{F}_2^n$ lead to subspace cosets and a combinatorial argument. Consequently, the continuous theorem neither proves nor directly implies Theorem 3.1.

7 Relation to earlier work

Exact support uncertainty. Donoho and Stark established the finite support-product inequality in the signal-recovery setting [DS89]. Meshulam and others developed sharper uncertainty inequalities that depend on the subgroup structure of the finite abelian group [Mes06]. Equality cases and related generalizations are treated explicitly by Bonami–Ghobber and Feng–Hollmann–Xiang [BG13, FHX19]; see also the survey of Wigderson–Wigderson [WW21]. These results supply the exact baseline: a character restricted to a subgroup coset is an extremizer. They do not by themselves give the leakage-tolerant quantitative conclusion of Theorem 3.1.

Miller's robust coefficient inequality. Miller [Mil26, Theorem 3.2 and Corollary 3.1] defines, for a nonzero function on an arbitrary finite abelian group, a uniformity coefficient $\nu(f)$ and a linearity coefficient $\eta(f)$. His corollary is

$$\frac{|\operatorname{supp} f|\,|\operatorname{supp} \widehat{f}|}{|G|}\,\nu(f)\eta(f) \geq 1.$$

Since both coefficients are at most one, a support product near $|G|$ forces them near one. Equality of ν means that $|f|$ is constant on its support; equality of η forces the support to be a subgroup coset. This is a broad and elegant robust diagnostic, valid for all finite abelian groups.

Theorem 3.1 is not a special case of that displayed inequality, nor does it subsume it. The hypotheses and outputs differ:

- Miller uses the literal supports of f and $\widehat{f}$. Theorem 3.1 uses arbitrary concentration sets S, T and remains informative when both literal supports are all of V.
- Miller controls scalar coefficients measuring magnitude uniformity and additive structure. Theorem 3.1 gives explicit symmetric-difference bounds for both complementary cosets.
- Theorem 3.1 also controls the full complex vector in squared ℓ_2 distance, including modulation and global phase.
- Miller works over arbitrary finite abelian groups. The present theorem uses the special exponent-two geometry of $\mathbb{F}_2^n$, including a four-point sign identity and a sharp small-doubling step.

The safest description is that the present result is a quantitative, noise-model-specific strengthening in the Boolean case, with a formal proof, rather than a new general uncertainty principle.

Continuous stability. The Gaussian conclusion for the Heisenberg variance inequality is the closest conceptual analogue, as discussed in Section 6. It belongs to a different branch of uncertainty theory: its notion of spread, symmetry group, extremizers, and proof mechanism all differ from the finite support problem. We include the comparison because it clarifies the term "robust inverse uncertainty," not because either result subsumes the other.

8 Lean formalization and audit surface

The formalization uses Lean 4.32.1 and Mathlib. It is organized so that an auditor need not read the implementation before understanding the claim.

File	Role
`Specification.lean`	Mathlib-only definitions of the cube, mass, Walsh transform, cosets, hypotheses, and conclusion.
`Theorem.lean`	The public theorem, its displayed mathematical docstring, and a one-line proof delegation.
`Internal/Bridge.lean`	Conversions between the public definitions and implementation definitions.
`Robust.lean`	Analytic reduction, set and projector estimates, phase normalization, and the final constants.
`Analytic.lean`	Parseval identities, restricted-Walsh residuals, mismatch counting, and the four-point argument.
`Combinatorial.lean`	Finite Markov, small doubling, and saturated almost orthogonality.
`Basic.lean`	Boolean Fourier and hidden-coset algebra.

The public declaration is:

```
theorem robust_inverse_uncertainty_phase
    (h : NearExtremizer epsilon delta q S T f) :
    exists H a b c, HiddenCosetApproximation q S T f H a b c
```

The project contains no proof placeholders, project-local axioms, opaque declarations, unsafe declarations, or uses of `native_decide`. The trusted base remains the Lean kernel, the imported Mathlib definitions and theorems, and the ordinary compiler/runtime assumptions needed to check the files [dMU21, The20].

9 Discussion

The theorem supports three conclusions. First, cosets are not an artifact of exact support: they persist under a concrete two-basis noise model. Second, the vector conclusion is stronger than a statement about sets; near saturation identifies a quantum state up to the expected translation, modulation, and global phase. Third, formalization is useful here because the argument combines Fourier normalization, additive combinatorics, projector algebra, and many explicit constants.

There are also clear limitations. The result is local, the dependence on the concentration loss is fourth-root, the constants are coarse, and the argument is specialized to exponent two. A sharper proof might improve the Young-inequality loss, avoid repeated factor-two estimates in the projector step, or replace mismatch rounding with a weighted stability argument. Extending the concentration-set theorem to general finite abelian groups would require a structural lemma adapted to their subgroup and phase geometry.

AI assistance disclosure. An AI system assisted with proof exploration, Lean implementation, expository drafting, and literature organization. The repository records the checkable formal artifact. Mathematical claims and decisions about authorship, novelty, and publication remain the responsibility of the human author.

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